

LeapYearCalculatorLib Unit Testing using NUnit

Introduction

This experiment explains how to write automated unit test cases for the LeapYearCalculatorLib project using the NUnit framework. The application checks whether a given year is a leap year and also validates whether the input year is within the allowed range.

Objectives

- Learn automated unit testing using NUnit.
- Understand the benefits of parameterized test cases.
- Verify leap year logic using multiple inputs.
- Use Assert.That() while following the Single Assertion Rule.

Implementation

Step 1: Create a Class Library project named LeapYearCalculatorLib.Tests in the same solution.

Step 2: Rename the default test class to LeapYearCalculatorTests.cs.

Step 3: Add the reference of LeapYearCalculatorLib to the test project.

Step 4: Install NUnit and NUnit3TestAdapter using NuGet Package Manager.

Step 5: Create parameterized test cases using the TestCase attribute.

Step 6: Add test cases for leap years and verify that the result is 1.

Step 7: Add test cases for non-leap years and verify that the result is 0.

Step 8: Add test cases for invalid years (less than 1753 or greater than 9999) and verify that the result is -1.

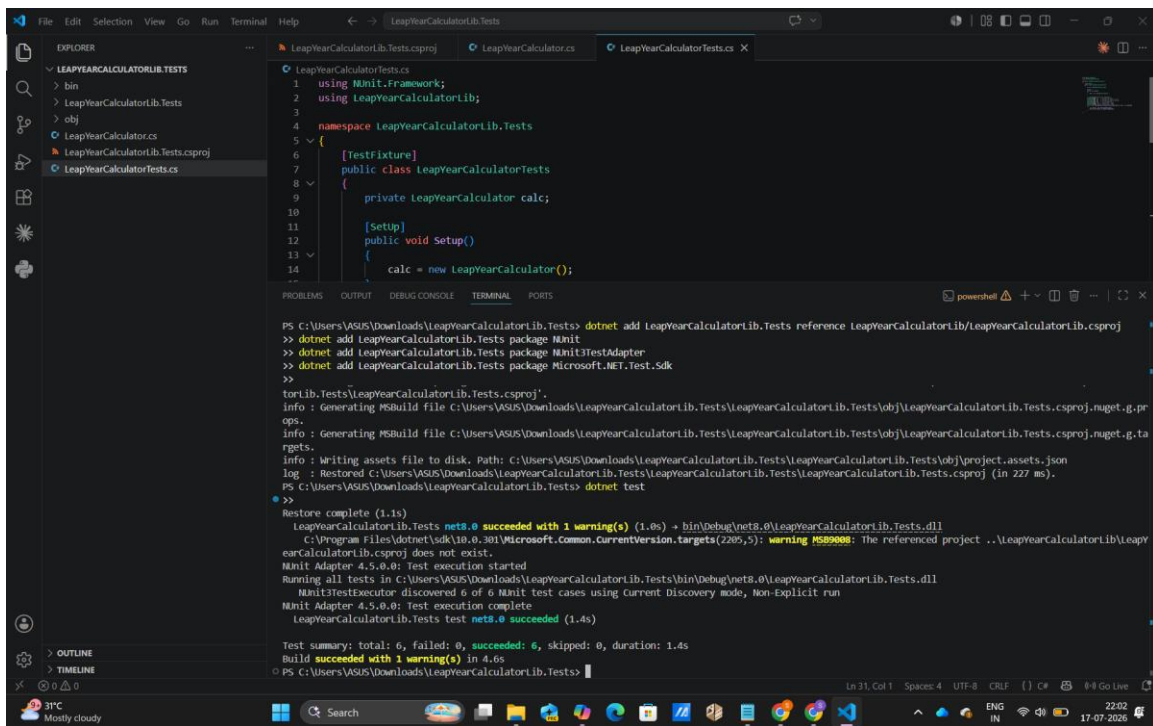
Step 9: Use Assert.That() to compare expected and actual results.

Step 10: Build the solution and run all the test cases.

Step 11: Modify the source code intentionally to break one test and rerun the tests to observe the failed result.

Execution

Paste the screenshot of the NUnit Test Explorer or terminal output here.



Expected Result

The application should return 1 for leap years, 0 for non-leap years, and -1 for invalid years outside the range 1753 to 9999. All parameterized test cases should pass.

Conclusion

This experiment helped in understanding parameterized unit testing using NUnit. Multiple input values were tested using a single test method, reducing duplicate code and improving readability.